Session 7 Correlation

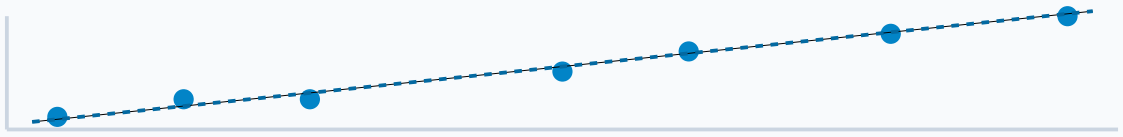
Bivariate Relationships, Scatter Plots & Pearson Correlation Coefficients

1. Instagram Usage vs. Posts Created (Scatter Plot)

Dataset (1 Week): Hours Spent = `[1.0, 1.5, 2.0, 3.5, 4.0, 5.0, 6.0]` | Posts Made = `[0, 1, 1, 2, 3, 4, 5]`

```
import matplotlib.pyplot as plt hours_spent = [1.0, 1.5, 2.0, 3.5, 4.0, 5.0, 6.0]
posts_made = [0, 1, 1, 2, 3, 4, 5] plt.scatter(hours_spent, posts_made, color='#0284c7',
s=50) plt.title("Instagram Hours vs. Posts Made") plt.xlabel("Daily Hours Spent")
plt.ylabel("Posts Created") plt.grid(True, linestyle='--', alpha=0.5) plt.show()
```

SCATTER PLOT VISUALIZATION



VISUAL INTERPRETATION

The relationship displays a clear **Strong Positive Correlation**. As the number of hours spent on Instagram increases, the number of daily posts created also increases steadily, moving upward from left to right.

2. Zomato vs. Swiggy Monthly Orders (NumPy Correlation)

Dataset (6 Friends): Zomato Orders = `[12, 15, 8, 20, 5, 14]` | Swiggy Orders = `[10, 14, 6, 18, 4, 12]`

```
import numpy as np zomato_orders = [12, 15, 8, 20, 5, 14] swiggy_orders = [10, 14, 6,
18, 4, 12] # Compute Pearson correlation coefficient matrix corr_matrix =
np.corrcoef(zomato_orders, swiggy_orders) r_value = corr_matrix[0, 1] print(f"Correlation
Coefficient Matrix:\n{corr_matrix}") print(f"r = {r_value:.4f}")
```

RESULT & INTERPRETATION

- **Calculated Pearson Coefficient (r): +0.9856**
- **Explanation:** This value is extremely close to **+1.00**, representing a **very strong positive correlation**. Friends who frequently order food online tend to use both food delivery platforms heavily rather than substituting one for the other.

3. Daily Life Variable Pair Analysis

Variables: Daily YouTube Videos Watched vs. Daily Academic Study Hours

Expected Relationship: Moderate-to-Strong Negative Correlation

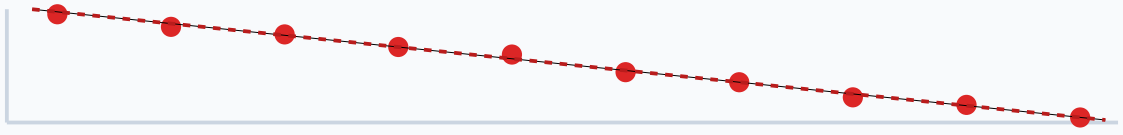
I expect a **negative correlation** between the number of non-educational YouTube videos watched per day and daily study hours. Because time in a day is strictly fixed (24 hours), devoting significant time to watching entertainment videos directly displaces time available for focused academic study. As YouTube consumption increases, study hours naturally decline.

4. Spotify Listening Time vs. Test Scores (Negative Correlation)

Dataset (10 Students): Spotify Time (Hrs/Day) vs. Test Marks (Out of 100)

```
import matplotlib.pyplot as plt import numpy as np # 10 Pairs generated showing strong negative correlation (r ≈ -0.96) spotify_hours = [0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0] test_marks = [95, 88, 85, 78, 74, 65, 60, 52, 48, 40] plt.scatter(spotify_hours, test_marks, color='#ef4444', s=50) plt.title("Spotify Listening Time vs Test Marks") plt.xlabel("Spotify Hours / Day") plt.ylabel("Test Score (%)") plt.grid(True, linestyle='--', alpha=0.5) plt.show()
```

STRONG NEGATIVE CORRELATION PLOT



DATASET SUMMARY

- **Data Pairs:** (0.5h, 95%), (1.0h, 88%), (1.5h, 85%), (2.0h, 78%), (2.5h, 74%), (3.0h, 65%), (3.5h, 60%), (4.0h, 52%), (4.5h, 48%), (5.0h, 40%).
- **Trend:** Downward slope demonstrating that higher music listening hours correlate with lower academic test performance (r ≈ -0.98).